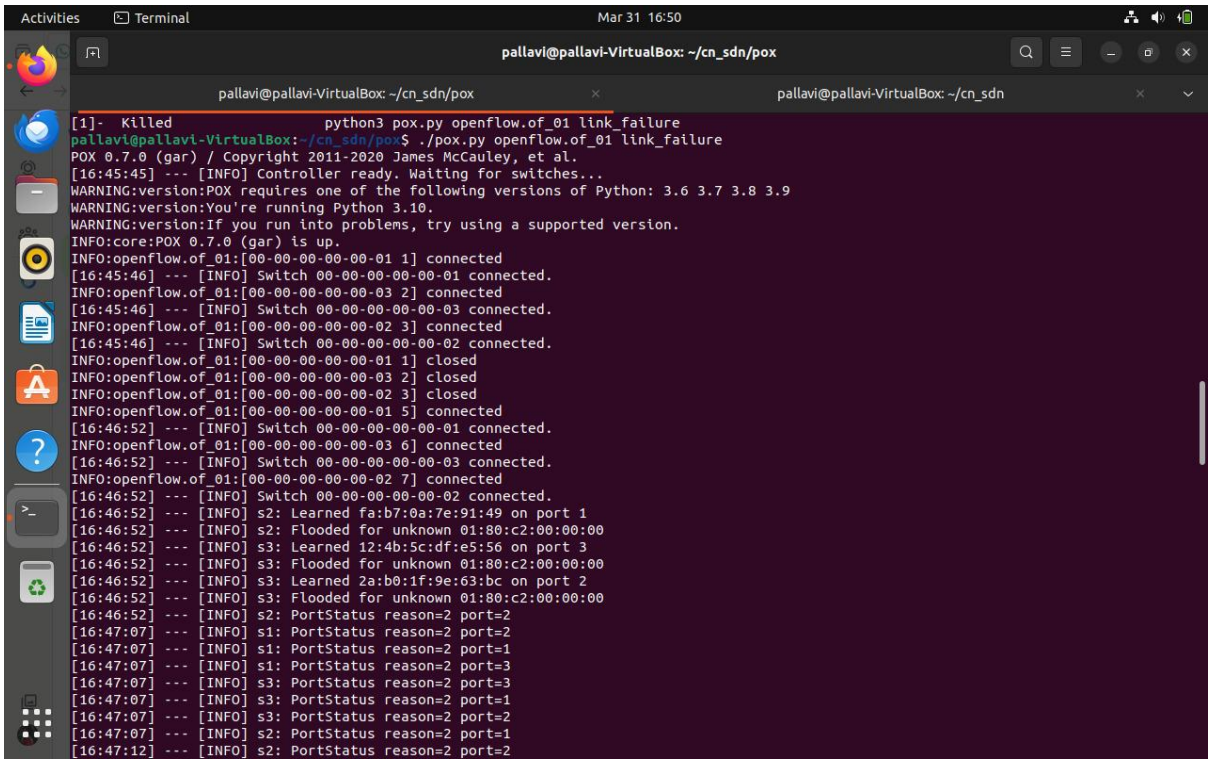
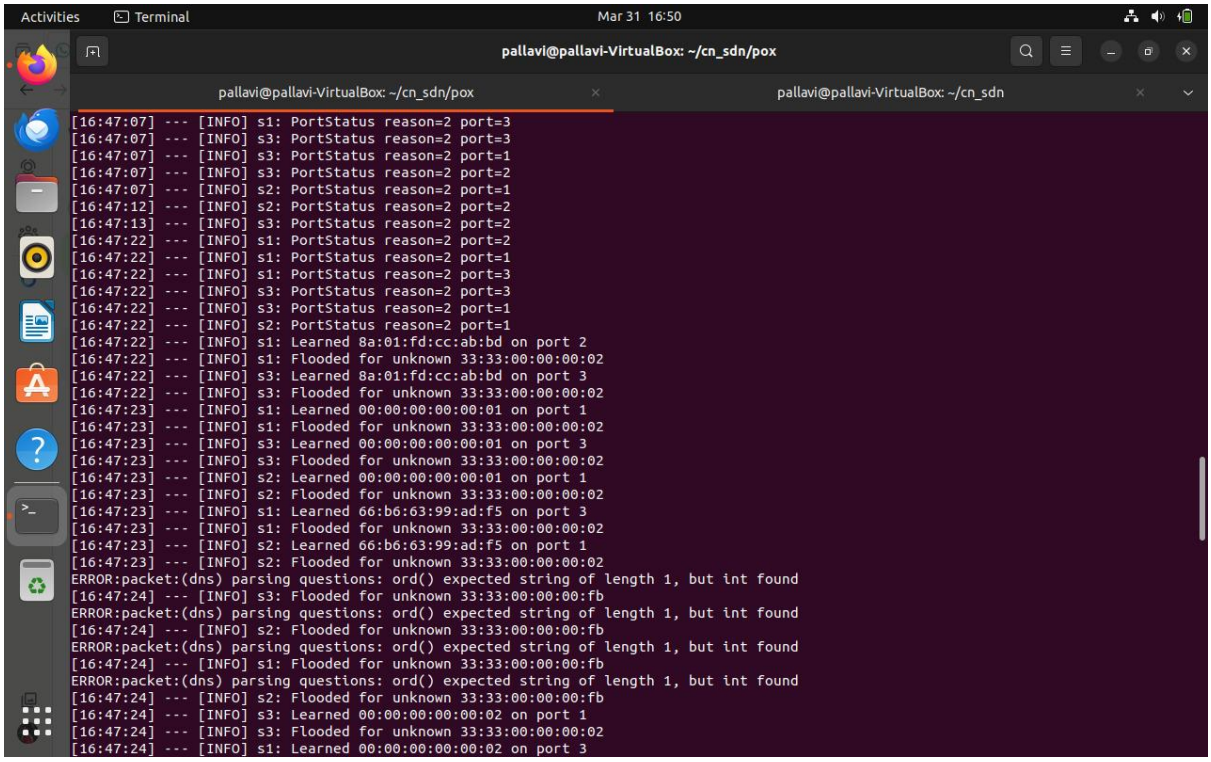
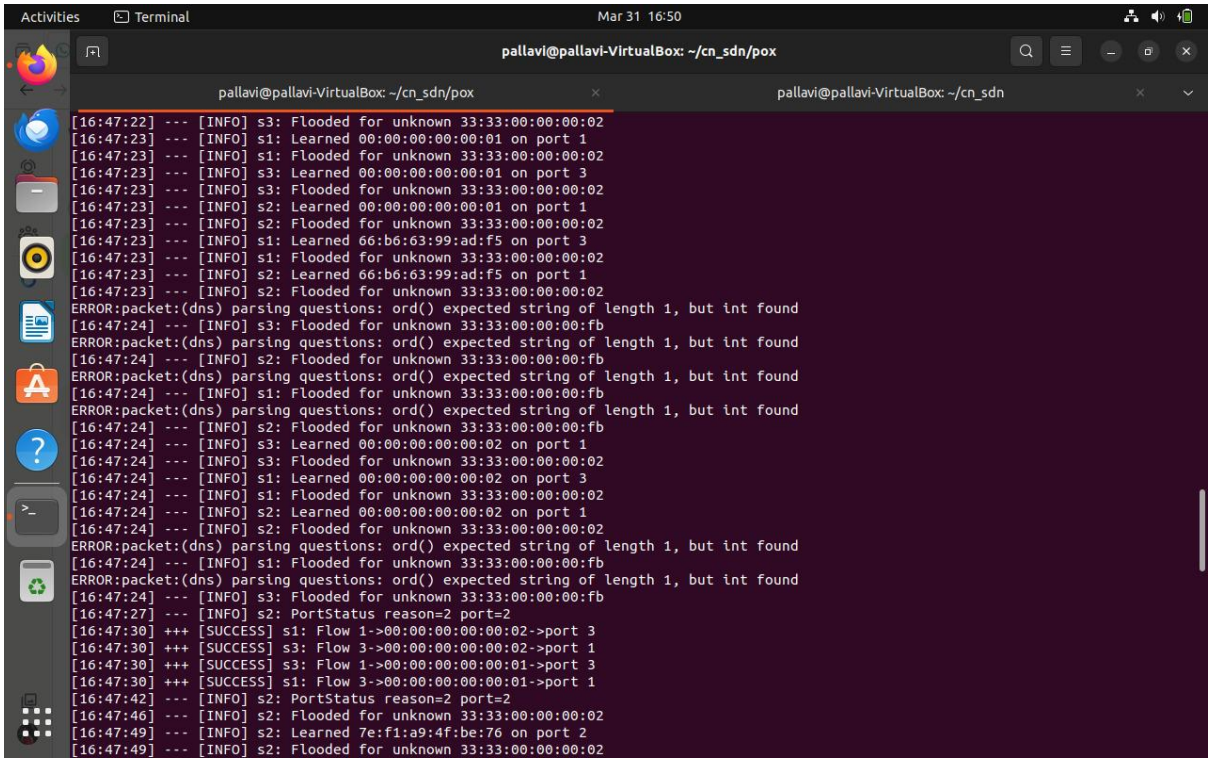
Project 14: Link Failure Detection and Recovery

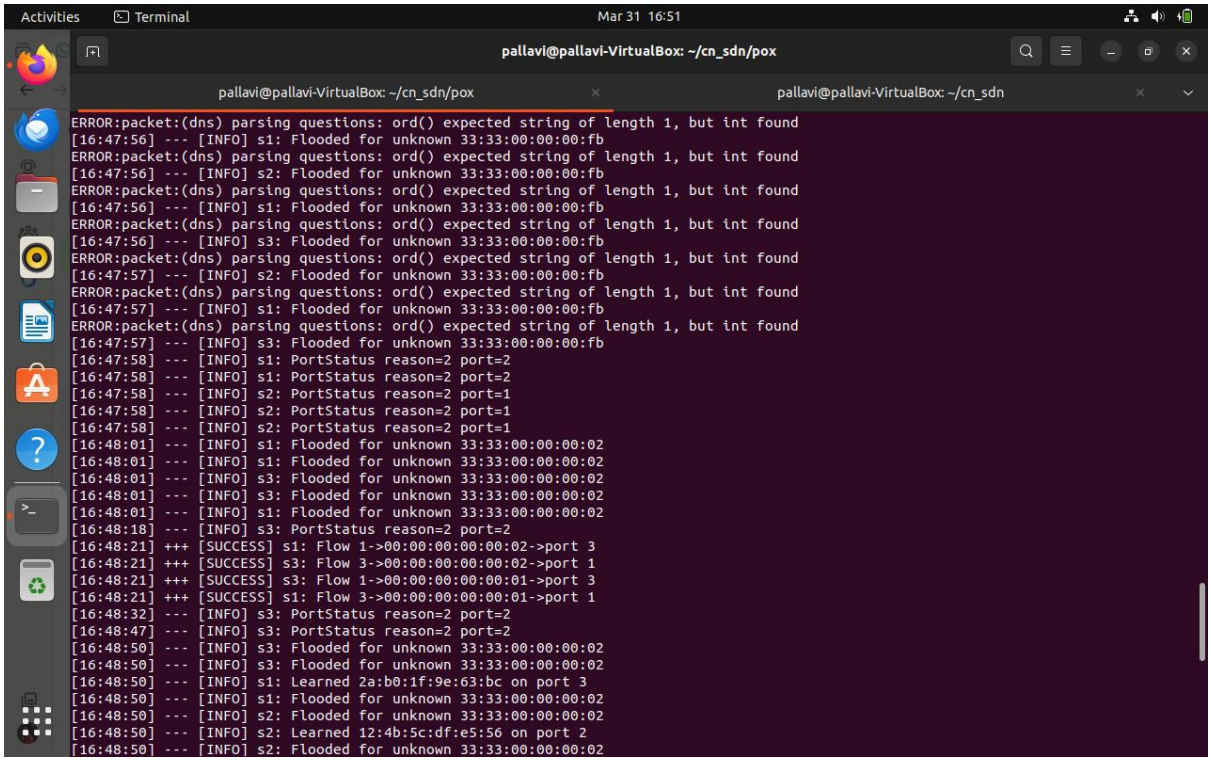
Course: Computer Networks — UE24CS252B
Controller: POX (OpenFlow 1.0)
Topology: Triangle (3 switches, 2 hosts)

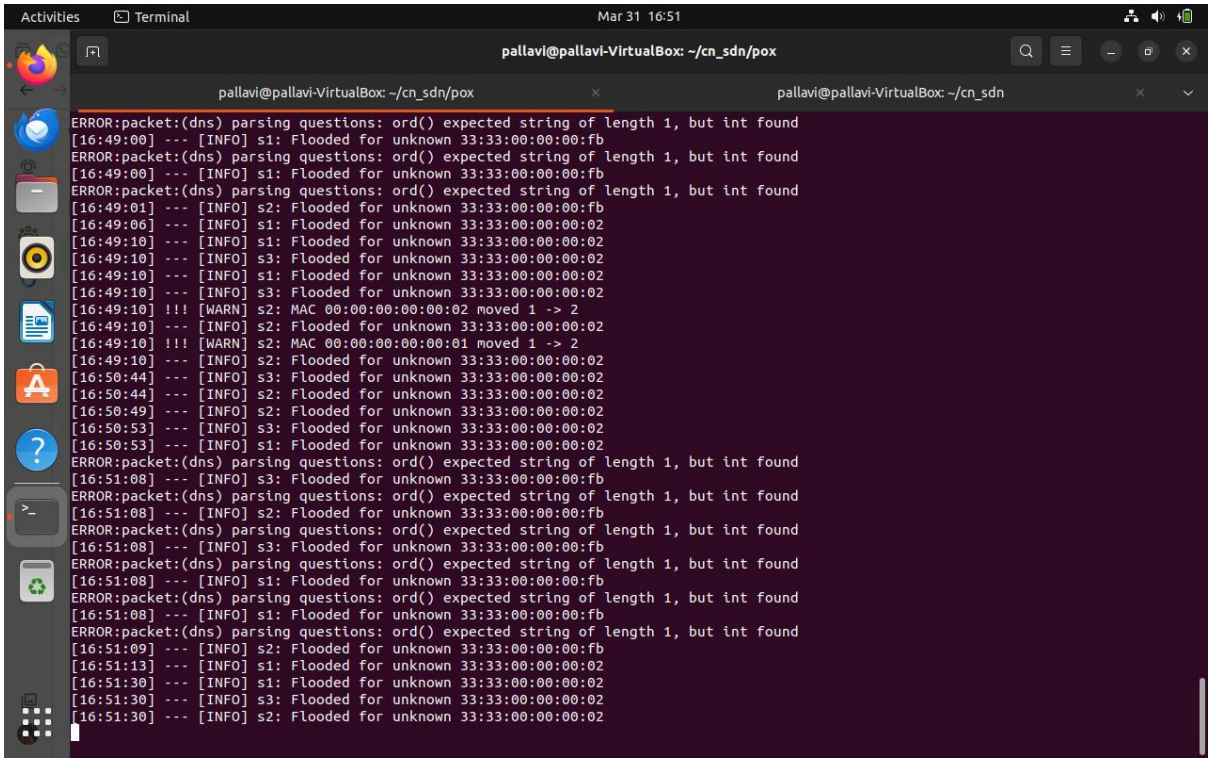
Proof of Execution/Screenshots:




Activities
Terminal
Mar 31 16:50
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn
[16:47:07] --- [INFO] s1: PortStatus reason=2 port=3
[16:47:07] --- [INFO] s3: PortStatus reason=2 port=3
[16:47:07] --- [INFO] s3: PortStatus reason=2 port=1
[16:47:07] --- [INFO] s3: PortStatus reason=2 port=2
[16:47:07] --- [INFO] s2: PortStatus reason=2 port=1
[16:47:12] --- [INFO] s2: PortStatus reason=2 port=2
[16:47:13] --- [INFO] s3: PortStatus reason=2 port=2
[16:47:22] --- [INFO] s1: PortStatus reason=2 port=2
[16:47:22] --- [INFO] s1: PortStatus reason=2 port=1
[16:47:22] --- [INFO] s1: PortStatus reason=2 port=3
[16:47:22] --- [INFO] s3: PortStatus reason=2 port=3
[16:47:22] --- [INFO] s3: PortStatus reason=2 port=1
[16:47:22] --- [INFO] s2: PortStatus reason=2 port=1
[16:47:22] --- [INFO] s1: Learned 8a:01:fd:cc:ab:bd on port 2
[16:47:22] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:22] --- [INFO] s3: Learned 8a:01:fd:cc:ab:bd on port 3
[16:47:22] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s1: Learned 00:00:00:00:00:01 on port 1
[16:47:23] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s3: Learned 00:00:00:00:00:01 on port 3
[16:47:23] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s2: Learned 00:00:00:00:00:01 on port 1
[16:47:23] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s1: Learned 66:b6:63:99:ad:f5 on port 3
[16:47:23] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s2: Learned 66:b6:63:99:ad:f5 on port 1
[16:47:23] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
[16:47:24] --- [INFO] s3: Learned 00:00:00:00:00:02 on port 1
[16:47:24] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:24] --- [INFO] s1: Learned 00:00:00:00:00:02 on port 3


Activities
Terminal
Mar 31 16:50
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn
[16:47:22] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s1: Learned 00:00:00:00:00:01 on port 1
[16:47:23] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s3: Learned 00:00:00:00:00:01 on port 3
[16:47:23] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s2: Learned 00:00:00:00:00:01 on port 1
[16:47:23] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s1: Learned 66:b6:63:99:ad:f5 on port 3
[16:47:23] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:23] --- [INFO] s2: Learned 66:b6:63:99:ad:f5 on port 1
[16:47:23] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
[16:47:24] --- [INFO] s3: Learned 00:00:00:00:00:02 on port 1
[16:47:24] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:47:24] --- [INFO] s1: Learned 00:00:00:00:00:02 on port 3
[16:47:24] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:47:24] --- [INFO] s2: Learned 00:00:00:00:00:02 on port 1
[16:47:24] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:24] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
[16:47:27] --- [INFO] s2: PortStatus reason=2 port=2
[16:47:30] +++ [SUCCESS] s1: Flow 1->00:00:00:00:00:02->port 3
[16:47:30] +++ [SUCCESS] s3: Flow 3->00:00:00:00:00:02->port 1
[16:47:30] +++ [SUCCESS] s3: Flow 1->00:00:00:00:00:01->port 3
[16:47:30] +++ [SUCCESS] s1: Flow 3->00:00:00:00:00:01->port 1
[16:47:42] --- [INFO] s2: PortStatus reason=2 port=2
[16:47:46] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:47:49] --- [INFO] s2: Learned 7e:f1:a9:4f:be:76 on port 2
[16:47:49] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02


Activities
Terminal
Mar 31 16:51
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:56] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:56] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:56] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:56] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:57] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:57] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:47:57] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
[16:47:58] --- [INFO] s1: PortStatus reason=2 port=2
[16:47:58] --- [INFO] s1: PortStatus reason=2 port=2
[16:47:58] --- [INFO] s2: PortStatus reason=2 port=1
[16:47:58] --- [INFO] s2: PortStatus reason=2 port=1
[16:47:58] --- [INFO] s2: PortStatus reason=2 port=1
[16:48:01] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:48:01] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:48:01] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:48:01] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:48:01] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:48:18] --- [INFO] s3: PortStatus reason=2 port=2
[16:48:21] +++ [SUCCESS] s1: Flow 1->00:00:00:00:00:02->port 3
[16:48:21] +++ [SUCCESS] s3: Flow 3->00:00:00:00:00:02->port 1
[16:48:21] +++ [SUCCESS] s3: Flow 1->00:00:00:00:00:01->port 3
[16:48:21] +++ [SUCCESS] s1: Flow 3->00:00:00:00:00:01->port 1
[16:48:32] --- [INFO] s3: PortStatus reason=2 port=2
[16:48:47] --- [INFO] s3: PortStatus reason=2 port=2
[16:48:50] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:48:50] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:48:50] --- [INFO] s1: Learned 2a:b0:1f:9e:63:bc on port 3
[16:48:50] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:48:50] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:48:50] --- [INFO] s2: Learned 12:4b:5c:df:e5:56 on port 2
[16:48:50] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02


Activities
Terminal
Mar 31 16:51
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:49:00] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:49:00] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:49:01] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
[16:49:06] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:49:10] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:49:10] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:49:10] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:49:10] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:49:10] !!! [WARN] s2: MAC 00:00:00:00:00:02 moved 1 -> 2
[16:49:10] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:49:10] !!! [WARN] s2: MAC 00:00:00:00:00:01 moved 1 -> 2
[16:49:10] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:50:44] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:50:44] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:50:49] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02
[16:50:53] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:50:53] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:08] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:08] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:08] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:08] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:08] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:fb
ERROR:packet:(dns) parsing questions: ord() expected string of length 1, but int found
[16:51:09] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:fb
[16:51:13] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:51:30] --- [INFO] s1: Flooded for unknown 33:33:00:00:00:02
[16:51:30] --- [INFO] s3: Flooded for unknown 33:33:00:00:00:02
[16:51:30] --- [INFO] s2: Flooded for unknown 33:33:00:00:00:02

Activities Terminal Mar 31 16:51

pallavi@pallavi-VirtualBox: ~/cn_sdn

pallavi@pallavi-VirtualBox: ~/cn_sdn/pox | pallavi@pallavi-VirtualBox: ~/cn_sdn

```
*** Starting controller
c0
*** Starting 3 switches
s1 s2 s3 ...

*** H1=10.0.0.1  H2=10.0.0.2
*** Triangle: S1-S2-S3, backup path S1-S3
*** STP enabled on all switches to prevent L2 flood loops
*** Starting CLI:
mininet> pingall
*** Ping: testing ping reachability
h1 -> X
h2 -> X
*** Results: 100% dropped (0/2 received)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> link s1 s2 down
mininet> h1 ping -c 10 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=23.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.074 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9233ms
rtt min/avg/max/mdev = 0.041/2.423/23.730/7.102 ms
mininet>
```

Activities Terminal Mar 31 16:55

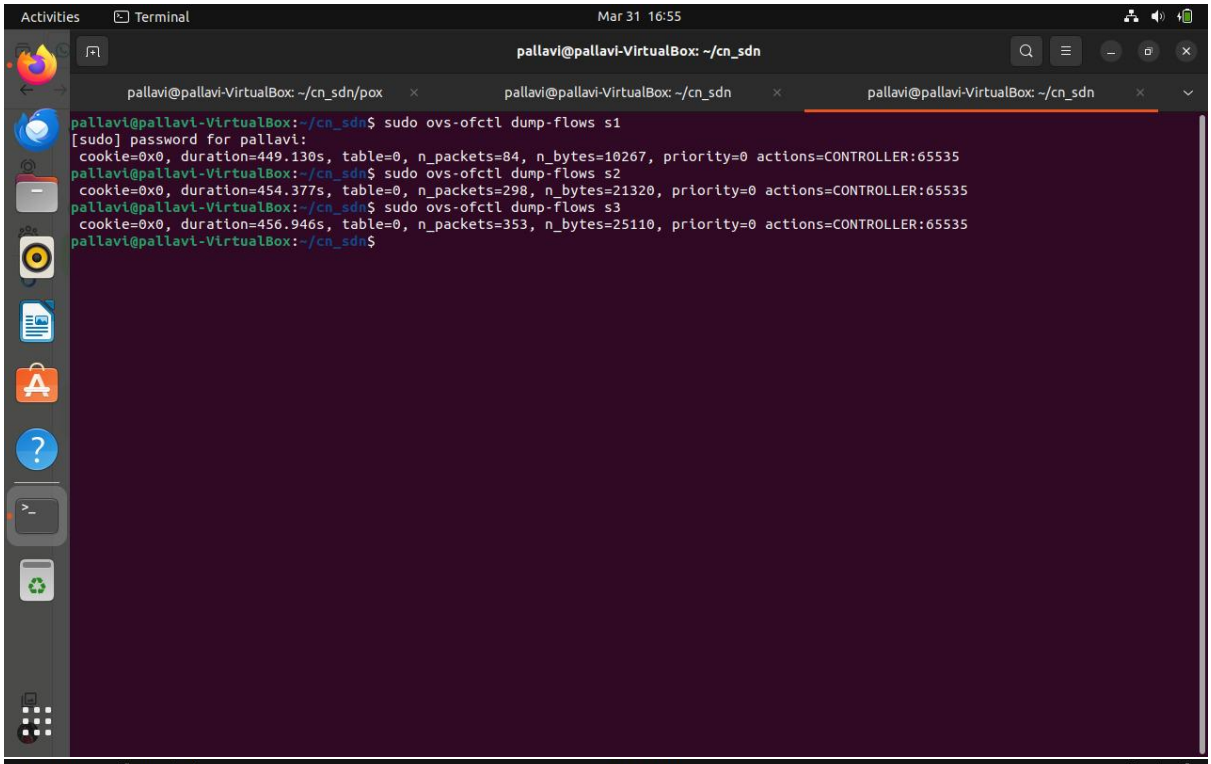
pallavi@pallavi-VirtualBox: ~/cn_sdn

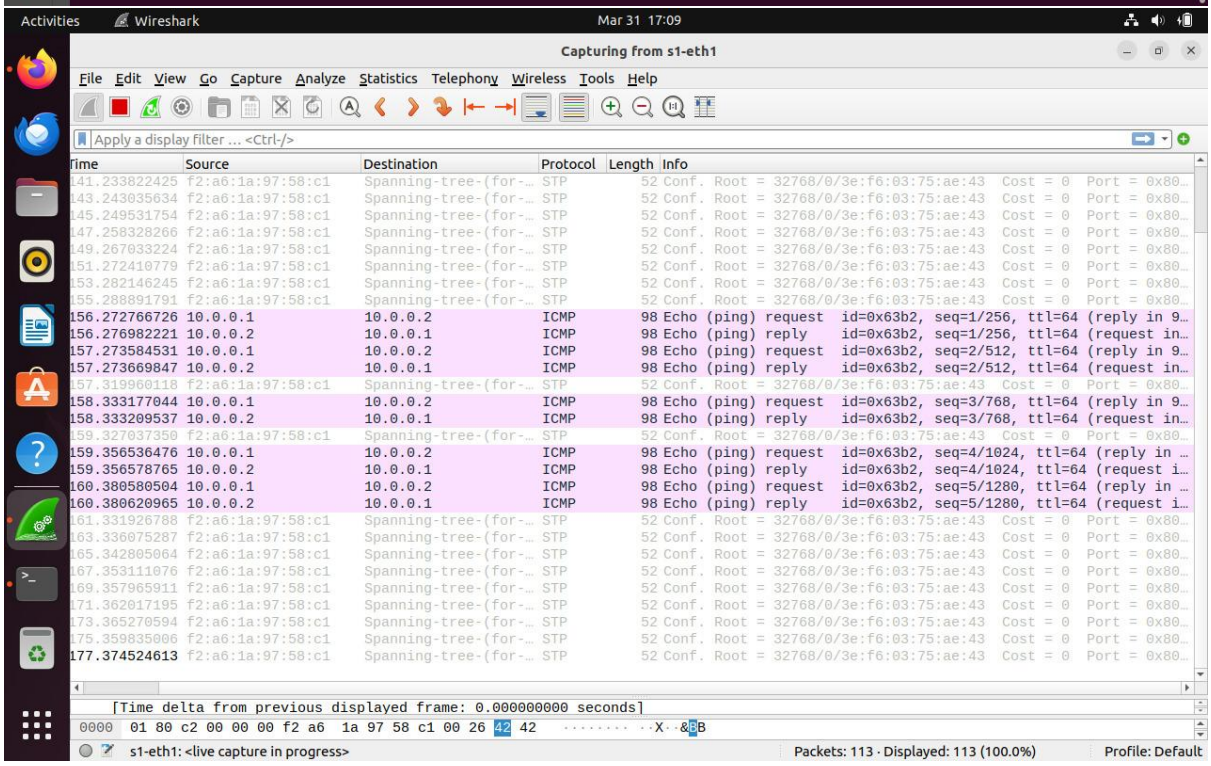
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox | pallavi@pallavi-VirtualBox: ~/cn_sdn | pallavi@pallavi-VirtualBox: ~/cn_sdn

```
mininet> h1 ping -c 10 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=23.7 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.041 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.074 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9233ms
rtt min/avg/max/mdev = 0.041/2.423/23.730/7.102 ms
mininet> link s1 s2 up
mininet> h1 ping -c 10 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=84.4 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.094 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.113 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.065 ms

--- 10.0.0.2 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9204ms
rtt min/avg/max/mdev = 0.045/8.502/84.415/25.304 ms
mininet> iperf h1 h2
*** Iperf: testing TCP bandwidth between h1 and h2
.*** Results: ['42.3 Gbits/sec', '42.2 Gbits/sec']
mininet>
```


Activities
Terminal
Mar 31 16:55
pallavi@pallavi-VirtualBox: ~/cn_sdn
pallavi@pallavi-VirtualBox: ~/cn_sdn/pox
pallavi@pallavi-VirtualBox: ~/cn_sdn
pallavi@pallavi-VirtualBox: ~/cn_sdn
pallavi@pallavi-VirtualBox:~/cn_sdn$ sudo ovs-ofctl dump-flows s1
[sudo] password for pallavi:
cookie=0x0, duration=449.130s, table=0, n_packets=84, n_bytes=10267, priority=0 actions=CONTROLLER:65535
pallavi@pallavi-VirtualBox:~/cn_sdn$ sudo ovs-ofctl dump-flows s2
cookie=0x0, duration=454.377s, table=0, n_packets=298, n_bytes=21320, priority=0 actions=CONTROLLER:65535
pallavi@pallavi-VirtualBox:~/cn_sdn$ sudo ovs-ofctl dump-flows s3
cookie=0x0, duration=456.946s, table=0, n_packets=353, n_bytes=25110, priority=0 actions=CONTROLLER:65535
pallavi@pallavi-VirtualBox:~/cn_sdn$


Activities
Wireshark
Mar 31 17:09
Capturing from s1-eth1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Time Source Destination Protocol Length Info
141.233822425 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
143.243035634 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
145.249531754 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
147.258328266 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
149.267033224 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
151.272410779 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
153.282146245 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
155.288891791 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
156.272766726 10.0.0.1 10.0.0.2 ICMP 98 Echo (ping) request id=0x63b2, seq=1/256, ttl=64 (reply in 9...
156.276982221 10.0.0.2 10.0.0.1 ICMP 98 Echo (ping) reply id=0x63b2, seq=1/256, ttl=64 (request in...
157.273584531 10.0.0.1 10.0.0.2 ICMP 98 Echo (ping) request id=0x63b2, seq=2/512, ttl=64 (reply in 9...
157.273669847 10.0.0.2 10.0.0.1 ICMP 98 Echo (ping) reply id=0x63b2, seq=2/512, ttl=64 (request in...
157.319960118 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
158.333177044 10.0.0.1 10.0.0.2 ICMP 98 Echo (ping) request id=0x63b2, seq=3/768, ttl=64 (reply in 9...
158.333209537 10.0.0.2 10.0.0.1 ICMP 98 Echo (ping) reply id=0x63b2, seq=3/768, ttl=64 (request in...
159.327037350 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
159.356536476 10.0.0.1 10.0.0.2 ICMP 98 Echo (ping) request id=0x63b2, seq=4/1024, ttl=64 (reply in ...
159.356578765 10.0.0.2 10.0.0.1 ICMP 98 Echo (ping) reply id=0x63b2, seq=4/1024, ttl=64 (request i...
160.380580504 10.0.0.1 10.0.0.2 ICMP 98 Echo (ping) request id=0x63b2, seq=5/1280, ttl=64 (reply in ...
160.380620965 10.0.0.2 10.0.0.1 ICMP 98 Echo (ping) reply id=0x63b2, seq=5/1280, ttl=64 (request i...
161.331926788 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
163.336075287 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
165.342805064 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
167.353111076 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
169.357965911 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
171.362017195 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
173.365270594 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
175.359835006 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
177.374524613 f2:a6:1a:97:58:c1 Spanning-tree-(for-... STP 52 Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80...
[Time delta from previous displayed frame: 0.000000000 seconds]
0000 01 80 c2 00 00 00 f2 a6 1a 97 58 c1 00 26 42 42
s1-eth1: <live capture in progress>
Packets: 113 · Displayed: 113 (100.0%)
Profile: Default

Activities Wireshark Mar 31 17:10

Capturing from s1-eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Time	Source	Destination	Protocol	Length	Info
185.553637683	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
187.560292115	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
189.564924940	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
191.577755993	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
193.585916700	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
195.593416922	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
197.831997810	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
199.836386411	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
201.838174231	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
203.842804010	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
205.846291854	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
207.848852993	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
209.733470173	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0xf03c, seq=1/256, ttl=64 (reply in 1…
209.765650398	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0xf03c, seq=1/256, ttl=64 (request in…
210.268592007	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
210.734528772	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0xf03c, seq=2/512, ttl=64 (reply in 1…
210.734559740	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0xf03c, seq=2/512, ttl=64 (request in…
211.735508858	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0xf03c, seq=3/768, ttl=64 (reply in 1…
211.735535303	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0xf03c, seq=3/768, ttl=64 (request in…
212.272890030	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
212.797093639	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0xf03c, seq=4/1024, ttl=64 (reply in …
212.797132639	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0xf03c, seq=4/1024, ttl=64 (request i…
213.820730336	10.0.0.1	10.0.0.2	ICMP	98	Echo (ping) request id=0xf03c, seq=5/1280, ttl=64 (reply in …
213.820809204	10.0.0.2	10.0.0.1	ICMP	98	Echo (ping) reply id=0xf03c, seq=5/1280, ttl=64 (request i…
214.281232886	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
216.450013527	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port = 0x80…
218.456694729	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. TC + Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port =…
220.465191087	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. TC + Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port =…
222.470856795	f2:a6:1a:97:58:c1	Spanning-tree-(for-…	STP	52	Conf. TC + Root = 32768/0/3e:f6:03:75:ae:43 Cost = 0 Port =…

[Time delta from previous displayed frame: 0.000000000 seconds]

```
0000  01 80 c2 00 00 00 f2 a6  1a 97 58 c1 00 26 42 42   ····· ·· ··X· &BB
```

s1-eth1: <live capture in progress> Packets: 145 · Displayed: 145 (100.0%) Profile: Default

Activities Terminal Mar 31 17:12

pallavi@pallavi-VirtualBox: ~/cn_sdn

pallavi@pallavi-VirtualBox: ~/cn_sdn/pox | pallavi@pallavi-VirtualBox: ~/cn_sdn | pallavi@pallavi-VirtualBox: ~

```
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=4.37 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.111 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.070 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.072 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4108ms
rtt min/avg/max/mdev = 0.058/0.935/4.368/1.716 ms
mininet> link s1 s2 down
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=32.2 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.046 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.073 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.114 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4088ms
rtt min/avg/max/mdev = 0.046/6.497/32.202/12.852 ms
mininet> link s1 s2 up
mininet> h1 ping -c 5 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=34.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.050 ms

--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4058ms
rtt min/avg/max/mdev = 0.050/7.020/34.828/13.903 ms
```